Four methods, two benchmarks — and a ranking that moves

Crowd-field reconstruction from partial robot observations · ATC, 7 held-out days

2026-09-07

What the four methods actually offer

Method	Reconstruction	Uncertainty	Where sigma-hat comes from
Senseiver	yes	no	—
DINCAE	yes	yes	part of the method (information form)
EnKF	yes	yes	part of the method (ensemble spread)
4DVarNet	yes	yes	ADDED BY US (2 designs; not in the paper)

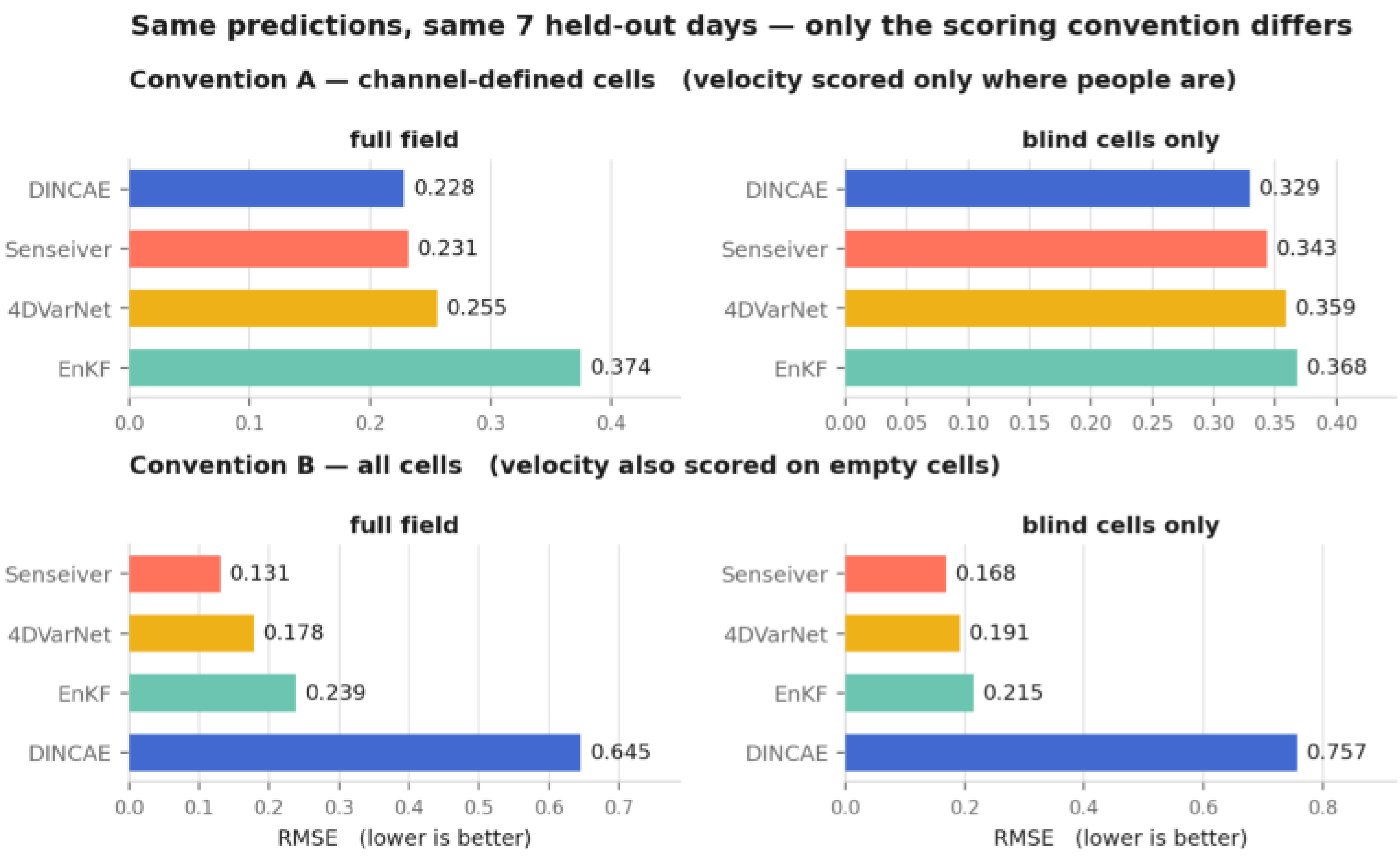
Accuracy compares 4 methods. Uncertainty compares 3 — and one of those three has a sigma-hat we added ourselves.

- Accuracy compares four methods. Uncertainty compares three — Senseiver has no sigma-hat at all.
- For DINCAE and the EnKF the uncertainty IS the method: remove the information form or the ensemble and you no longer have that method.
- For 4DVarNet we added it. Two designs, neither in the original paper.
  - So: four reconstruction methods, plus two 4DVarNet uncertainty designs that are ours, not the paper's.

Hypothesis

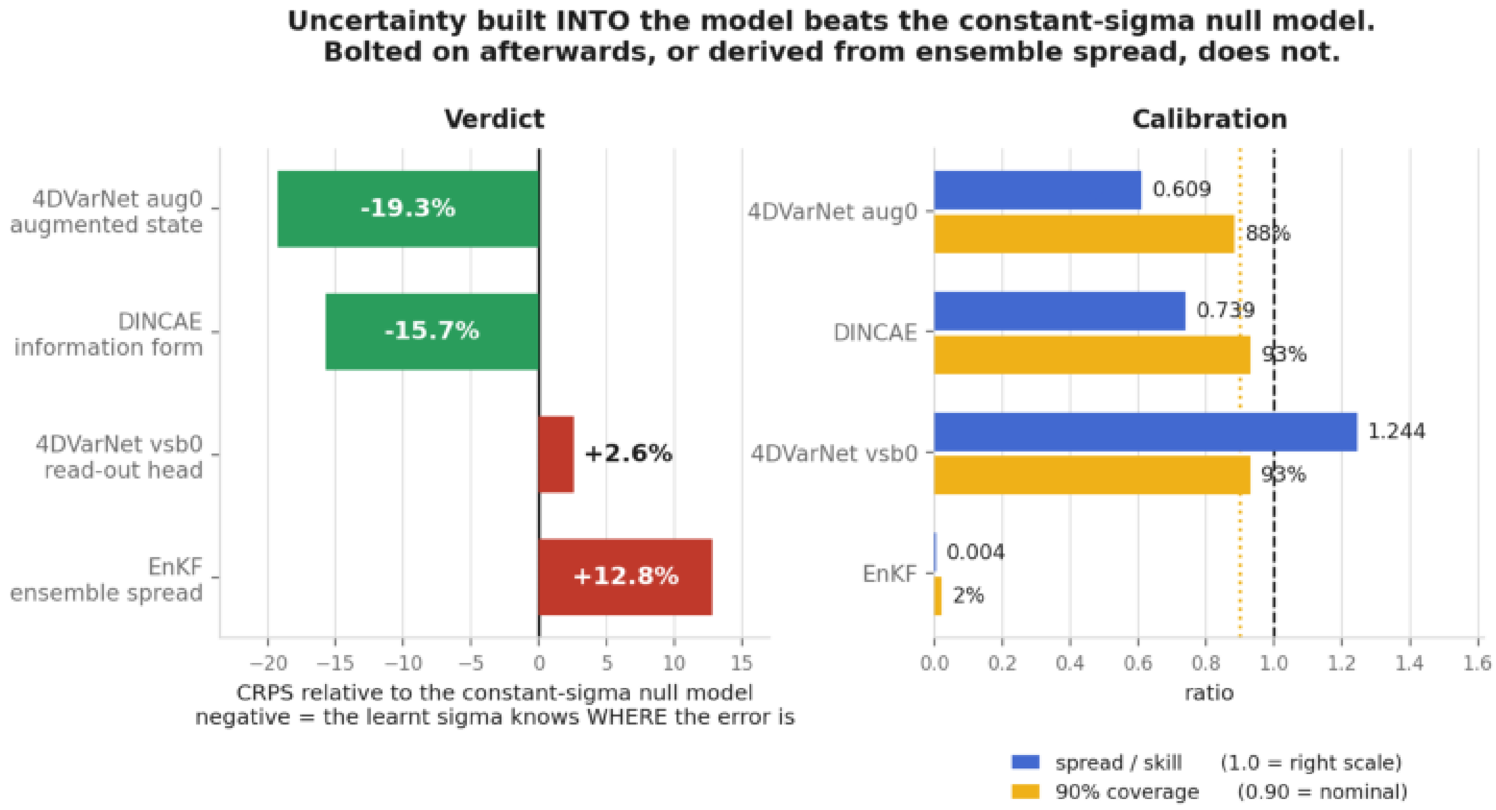
- The ranking of these four methods is decided by a scoring convention that most papers do not state.
- Two independent tests, both on the same 7 held-out days:
  - accuracy — which cells count
  - uncertainty — which cells count
- If the hypothesis is wrong, the ordering survives both. It survives neither.

Test 1 — accuracy: the same predictions, scored two ways



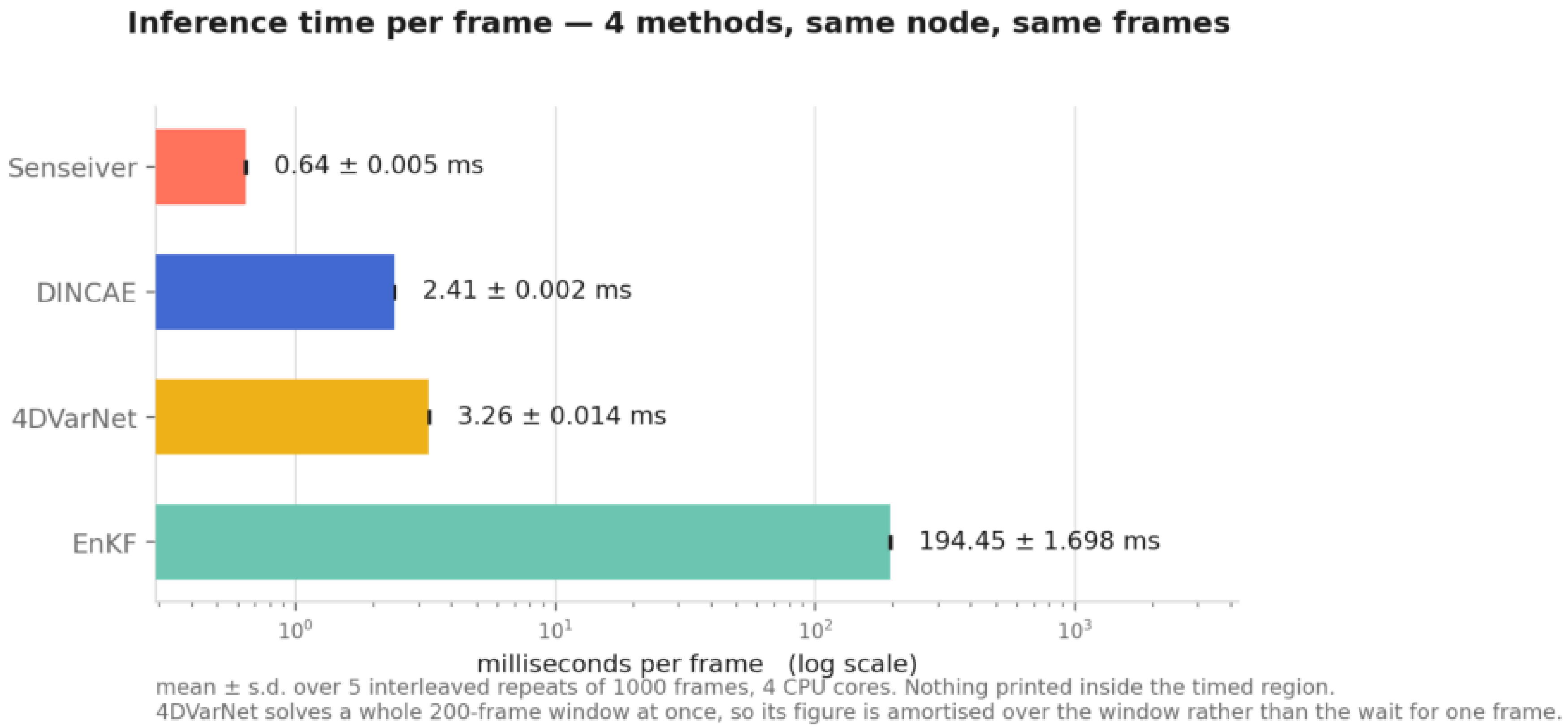
- DINCAE is first under one convention and last under the other — 4.5x worse than Senseiver. Nothing about the model changed.
- Cause: DINCAE only defines velocity where density > 0. 88.4% of blind velocity cells are empty, and on those the truth is exactly 0 (measured: 0.000% non-zero).
- Convention A is the defensible one — “velocity” on an empty cell is not a physical quantity.
- Left column scores the full field, right column the blind cells alone. Observed cells are handed to the model, so the right column is the reconstruction task.

Test 2 — uncertainty: what makes a sigma-hat useful



- Judged against a null model: replace every cell's sigma-hat with one constant, that split's own RMSE. It knows how big the error is on average and nothing about WHERE — and it scores a perfect 1.00 spread/skill for free.
- Uncertainty built into the model beats it. Bolted on afterwards, or read off ensemble spread, does not.
  - This ordering is the one thing in the deck that does NOT move with the convention.

Action item — inference time, per frame



- You were right that last week's figure was total run time, not inference time. This is per frame: mean ± s.d. over 5 interleaved repeats, all four methods on the same node over the same frames, with nothing printed inside the timed region.
- Senseiver is the fastest by a wide margin — and it is also the most accurate under Convention B.
  - Caveat on the 4DVarNet bar: it solves a whole 200-frame window at once, so its 2.74 ms is amortised over the window, not the wait for one frame.

The ablation that settles it

- Same DINCAE architecture. Same 7 days. Same validation-picked checkpoint rule. The only change: supervise every cell instead of only the defined ones.
  - all-cells blind RMSE: 0.757 -> 0.173 (-77%)
  - defined blind RMSE: 0.329 -> 0.349 (+5.9%)
- Last place, 4.5x behind, becomes second — by changing the loss mask, not the model. The gap under convention B was never a capability gap.
- And the 5.9% it loses under convention A says the information form is genuinely part of why DINCAE wins there — not just its architecture.